



COMPUTER NETWORKS

Computer Networks and the Internet

Team Networks

Department of Computer Science and Engineering

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Unit – 1 Computer Networks and the Internet

1.1 Introduction to Computer Networks

1.2 What is the Internet?

- A nuts-and-bolts and Services description, Protocol

1.3 Network edge

- End systems, Access networks, Physical media

1.4 Network core

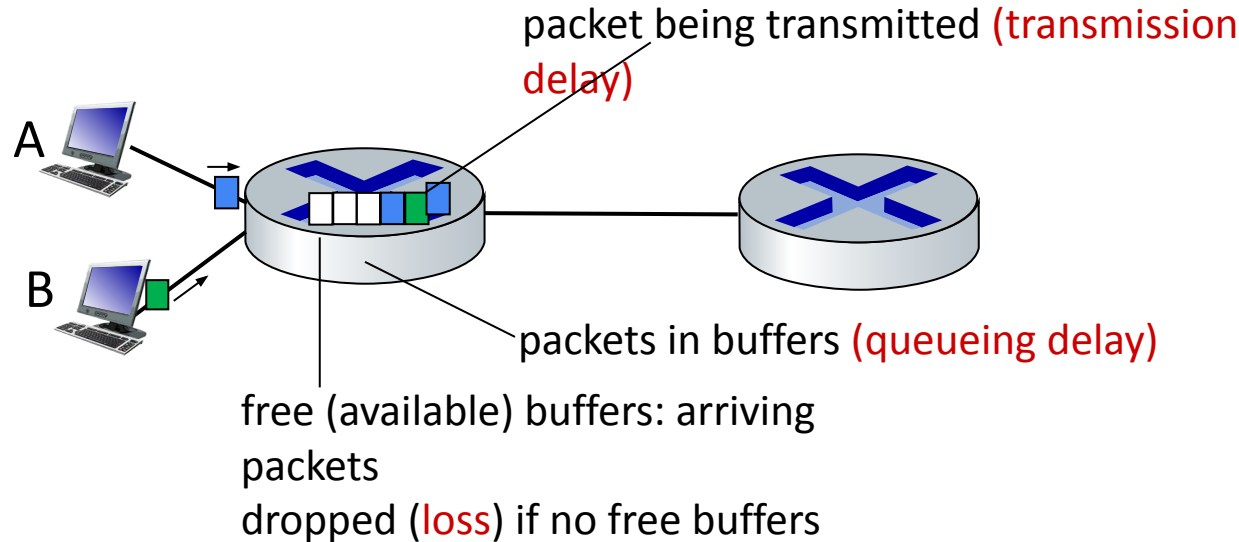
- Packet switching, Circuit switching, Network structure

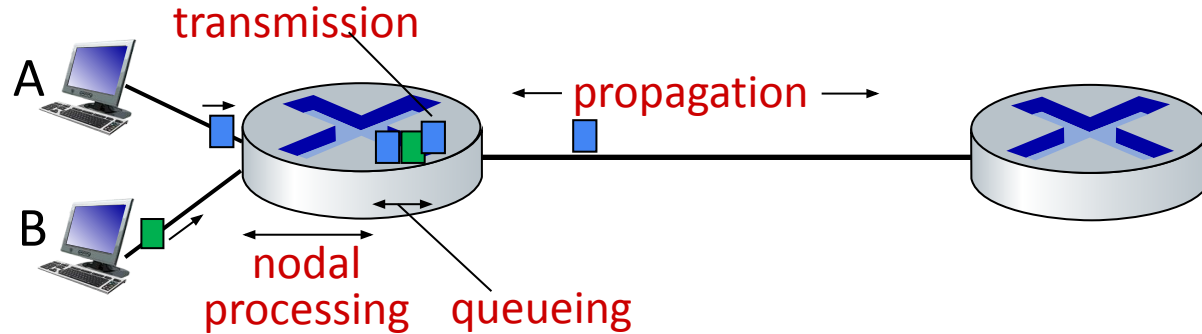
1.5 Delay, Loss & Throughput in networks

How do packet loss and delay occurs?

packets *queue* in router buffers

- packets queue, wait for turn
- arrival rate to link (temporarily) exceeds output link capacity: packet loss





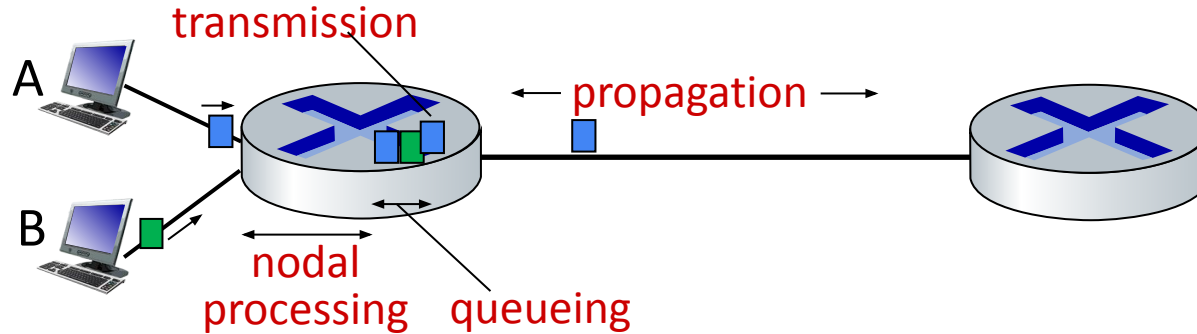
$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{proc} : nodal processing

- check bit errors
- determine output link
- typically < msec

d_{queue} : queueing delay

- time waiting at output link for transmission
- depends on congestion level of router
- microseconds to milliseconds



* Check out the online interactive exercises:
http://gaia.cs.umass.edu/kurose_ross

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{trans} : transmission delay:

- L : packet length (bits)
- R : link transmission rate (bps)

$$d_{\text{trans}} = L/R$$

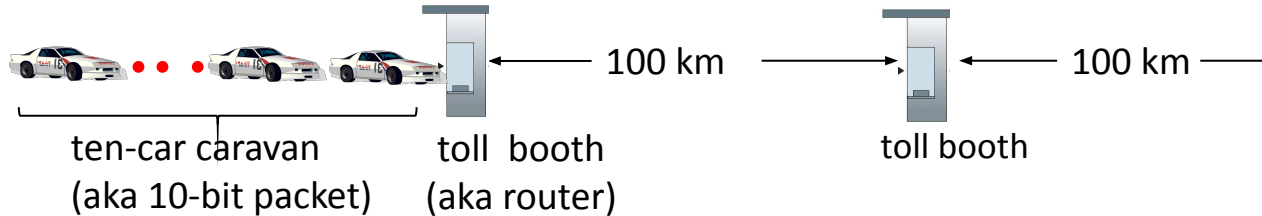
d_{prop} : propagation delay:

- d : length of physical link
- s : propagation speed ($\sim 2 \times 10^8$ m/sec)

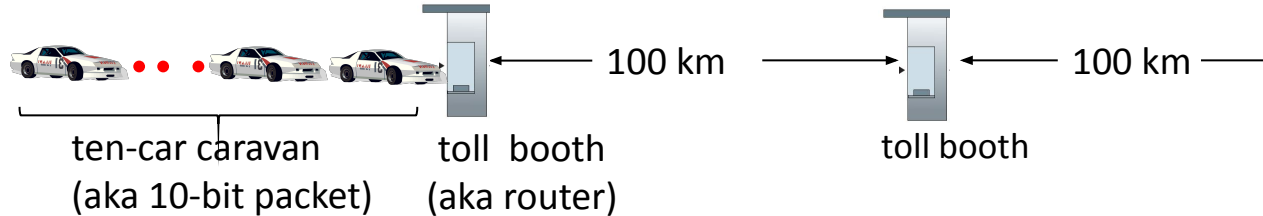
$$d_{\text{prop}} = d/s$$

d_{trans} and d_{prop}
very different

Transmission Delay	Propagation Delay
Time required for the router to push out the packet.	Time it takes a bit to propagate from one router to the next.
A function of the packet's length and the transmission rate of the link.	A function of the distance between the two routers.
$d_{trans} = L/R$	$d_{prop} = d/s$
Nothing to do with the distance between the two routers.	Nothing to do with the packet's length or the transmission rate of the link.



- cars “propagate” at 100 km/hr
- toll booth takes 12 sec to service car (bit transmission time)
- car ~ bit; caravan ~ packet
- **Q: How long until caravan is lined up before 2nd toll booth?**
- time to “push” entire caravan through toll booth onto highway = $12 \times 10 = 120$ sec
- time for last car to propagate from 1st to 2nd toll booth:
 $100\text{km} / (100\text{km/hr}) = 1$ hr
- **A: 62 minutes**



- suppose cars now “propagate” at 1000 km/hr
- and suppose toll booth now takes one min to service a car
- **Q: Will cars arrive to 2nd booth before all cars serviced at first booth?**

A: Yes! after 7 min, first car arrives at second booth; three cars still at first booth

Unlike other delays (d_{proc} , d_{trans} , d_{prop}), d_{queue} is interesting.

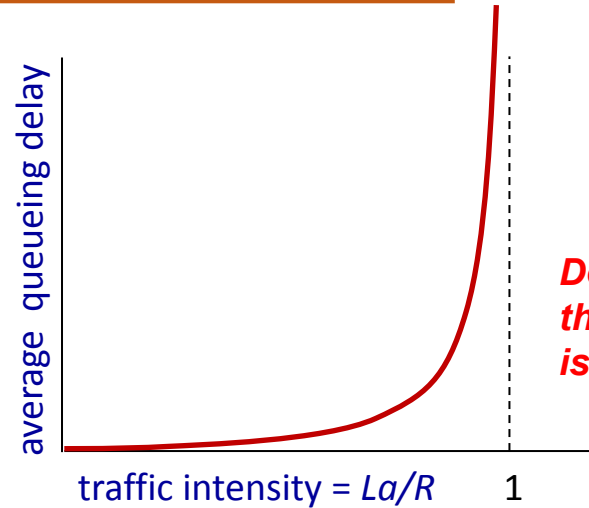
- Can vary from packet to packet.
- Characterize d_{queue} -> average, variance, probability that it exceeds some specified value.

When is the queuing delay large and when is it insignificant?

- Rate at which traffic arrives at the queue,
- Transmission rate of the link,
- Nature of the arriving traffic – periodically or in bursts

- R : link bandwidth (bps)
- L : packet length (bits)
- a : average packet arrival rate (pps)
- La : avg. rate at which bits arrive at the queue
- $La/R > 1$: more “work” arriving is more than can be serviced - average delay infinite!
- $La/R \leq 1$: nature of arriving traffic
- $La/R \sim 0$: avg. queueing delay small

$La/R > 1$: Average rate at which bits arrive at the queue exceeds the rate at which the bits can be transmitted from the queue.

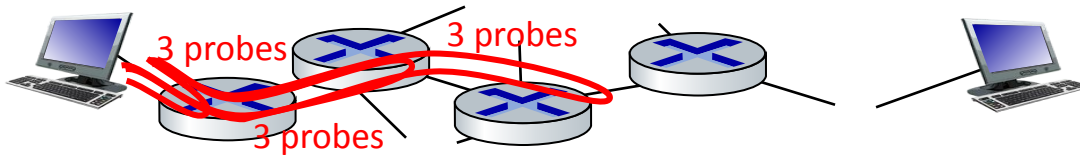


Design your system so that the traffic intensity is no greater than 1.



$La/R \rightarrow$

- what do “real” Internet delay & loss look like?
- **traceroute** program: provides delay measurement from source to router along end-end Internet path towards destination. For all i :
 - sends three packets that will reach router i on path towards destination (with time-to-live field value of i)
 - router i will return packets to sender
 - sender measures time interval between transmission and reply



traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu

```
1 cs-gw (128.119.240.254) 1 ms 1 ms 2 ms
2 border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145) 1 ms 1 ms 2 ms
3 cht-vbns.gw.umass.edu (128.119.3.130) 6 ms 5 ms 5 ms
4 jn1-at1-0-0-19.wor.vbns.net (204.147.132.129) 16 ms 11 ms 13 ms
5 jn1-so7-0-0-0.wae.vbns.net (204.147.136.136) 21 ms 18 ms 18 ms
6 abilene-vbns.abilene.ucaid.edu (198.32.11.9) 22 ms 18 ms 22 ms
7 nycm-wash.abilene.ucaid.edu (198.32.8.46) 22 ms 22 ms 22 ms
8 62.40.103.253 (62.40.103.253) 104 ms 109 ms 106 ms
9 de2-1.de1.de.geant.net (62.40.96.129) 109 ms 102 ms 104 ms
10 de.fr1.fr.geant.net (62.40.96.50) 113 ms 121 ms 114 ms
11 renater-gw.fr1.fr.geant.net (62.40.103.54) 112 ms 114 ms 112 ms
12 nio-n2.cssi.renater.fr (193.51.206.13) 111 ms 114 ms 116 ms
13 nice.cssi.renater.fr (195.220.98.102) 123 ms 125 ms 124 ms
14 r3t2-nice.cssi.renater.fr (195.220.98.110) 126 ms 126 ms 124 ms
15 eurecom-valbonne.r3t2.ft.net (193.48.50.54) 135 ms 128 ms 133 ms
16 194.214.211.25 (194.214.211.25) 126 ms 128 ms 126 ms
17 * * *
18 * * *
```

3 delay measurements
to border1-rt-fa5-1-0.gw.umass.edu

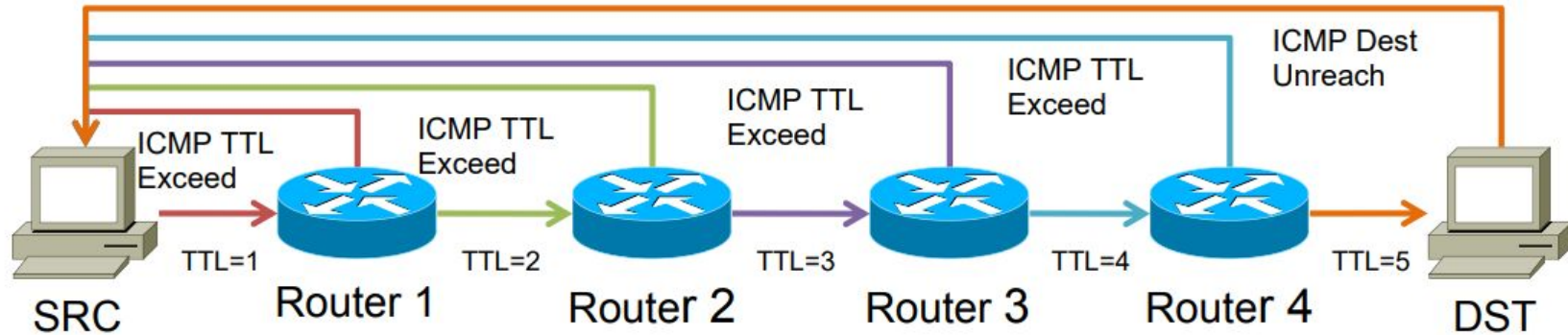
trans-oceanic link

looks like delays
decrease! Why?

* means no response (probe lost, router not replying)

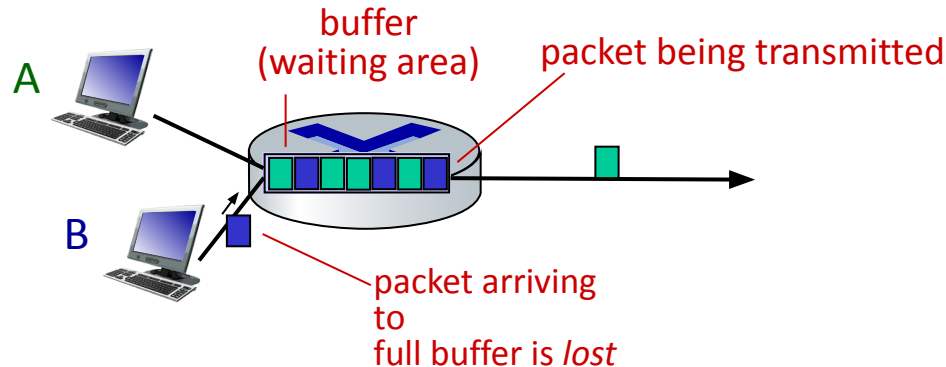
```
19 fantasia.eurecom.fr (193.55.113.142) 132 ms 128 ms 136 ms
```

* Do some traceroutes from exotic countries at



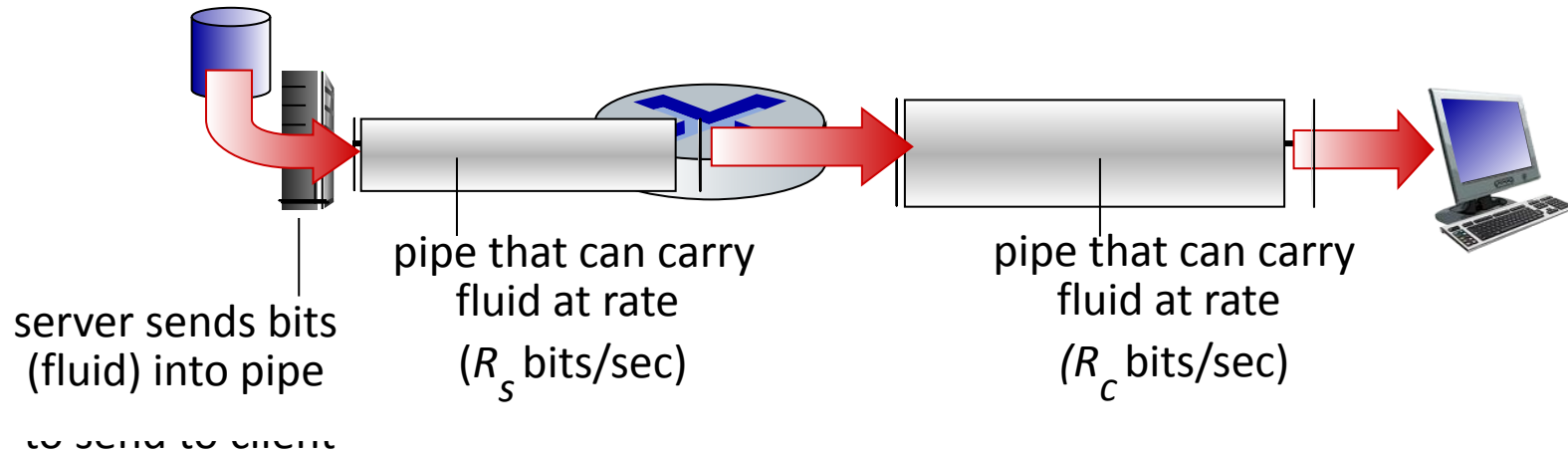
Refer RFC 1393, **Traceroute Using an IP Option**
Don't Trust Traceroute (Completely)

- queue (aka buffer) preceding link in buffer has finite capacity
- packet arriving to full queue dropped (aka lost)
- lost packet may be retransmitted by previous node, by source end system, or not at all

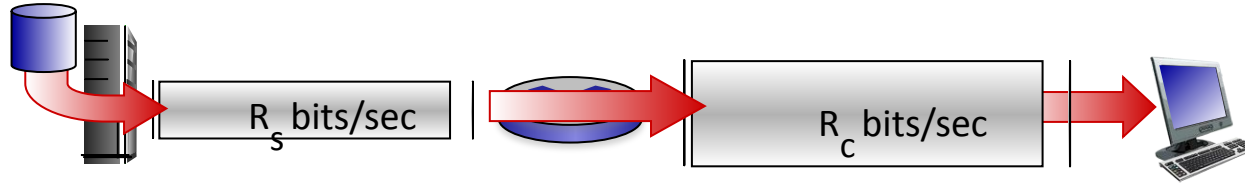


* Check out the Java applet for an interactive animation on queuing and loss

- *throughput*: rate (bits/time unit) at which bits are being sent from sender to receiver
 - *instantaneous*: rate at given point in time
 - *average*: rate over longer period of time

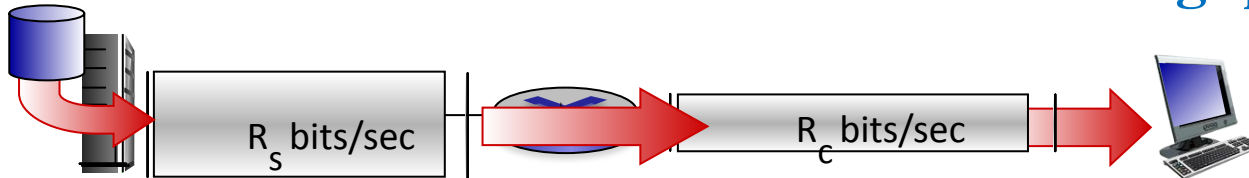


$R_s < R_c$ What is average end-end throughput?



$R_s > R_c$ What is average end-end throughput?

$$\text{Throughput} = \min\{R_s, R_c\}$$



bottleneck link

link on end-end path that constrains end-end throughput.

Throughput – Numerical Example

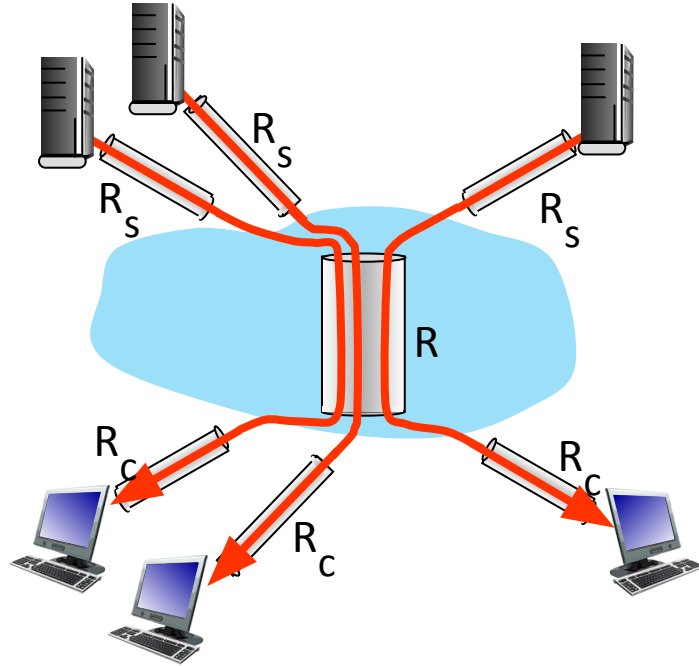
- Suppose you are downloading an MP3 file of $F = 32$ million bits.
- The server has a transmission rate of $R_s = 2$ Mbps and you have an access link of $R_c = 1$ Mbps.
- What is the time needed to transfer the file?

Let's work it out!

Solution:

- 32 seconds!





10 connections (fairly) share
backbone bottleneck link R bits/sec

- per-connection end-end throughput: $\min(R_c, R_s, R/10)$
- in practice: R_c or R_s is often bottleneck

- Suppose $R_s = 2$ Mbps, $R_c = 1$ Mbps, $R = 5$ Mbps
- 10 clients from 10 servers = 10 downloads

End-to-end throughput for
each download is now
reduced to 500 kbps.

* Check out the online interactive
exercises for more examples:
http://gaia.cs.umass.edu/kurose_ross/





THANK YOU

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